

Nilendu Ganguli

 GitHub  LinkedIn  nilendu.ganguli@gmail.com  8697370231

EXPERIENCE

CITI | DATA ENGINEER

Dec 2022 – Current | Pune, India

- Engaged in development of an **Event-driven Multimodal** ingestion pipeline for KYC documents, combining **Page-Index** and **B-Tree based Vectorized Micro-knowledge Graph** enhanced by **Hybrid Search Routing** and **Graph-Vector Synchronization**.
- Architected a **10-node LangGraph agentic pipeline** converting natural language to validated Oracle SQL, featuring a **custom in-memory Property Graph** (replacing Neo4j), **7-tool Autonomous Entity Extraction**, 6-check SQL validation with retry loops, supporting **1000+ tables** and employing **Inter-agent Communication** exhibiting **Natural Mimicry of business and tech-side interactions**.
- Engineered a **CPU-only, production-grade Enterprise RAG** pipeline ingesting 1TB/hour across 7 Python microservices with **hybrid HNSW+BM25+RRF retrieval**, **ONNX BERT INT8 embeddings**, and OpenShift autoscaling.

CITI | SOFTWARE DEVELOPER 2

Aug 2024 – Nov 2025 | Pune, India

- Engineered Solutions for KYC-AI Transformation , a system for generating **disposition summary and risk category assignment for Member Entities** across multiple Regions using **LLMs paired with workflows tailored to CITI business units**.
- This system leverages both **deterministic chains and multi-agent systems**, to process over **290 different types of document categories** defined under KYC taxonomy. This system aims to **reduce document processing time by 70%** than current operations efforts.
- Delivered multiple **backend microservices, micro-batch processing pipelines** to help in the KYC Migration Effort. As per business requirements, used **Fast API apps paired with Uvicorn and Gunicorn Servers to create Restful APIs** to fulfill Technical expectations (caters to a wide variety of ASGI and WSGI apps).
- Impact: Risk of Non-Compliance: Introduced much needed leanness in legacy KYC. Increased throughput by 6X without altering resource utilization. Yielded a **reduction in Response time from 230ms to 110 ms**.

CITI | SOFTWARE DEVELOPER

Jul 2022 – Aug 2024 | Pune, India

- Launched **KYC Document Processing and Archival Service** in North America, that eliminated vendor dependency for KYC and laid foundation for implementation of **KYC 2.0 in NAM and Europe** .
- Created **microservices to support CRUD operations** for KYC 1.0 compatibility and **implemented Luigi and Apache NIFI workflows** to enhance KYC 2.0, thereby processing upwards of **100K KYC Documents per day** and resulting in data and service availability for downstream KYC Services.
- Impact: **Completely automated KYC document processing**(mostly PDFs), storage and indexing workflows. **Reducing Client on-boarding time from 14 days to 2 days**.

INTERNSHIPS

CITI | SOFTWARE DEVELOPER

Jun 2021 - Jul 2021 | Pune, India

- Designed and Implemented a **BigData** framework for **Global Data Purge Policy(Europe)** on **KYC Production Data Warehouse**.
- Created **Apache Spark and Kafka Pipeline** for implementation of GDPR policy on Archival data for **CitiBank Member Entity** amounting to uncompressed **size of 10 TeraByte**.
- Deployed and tested the distributed system framework in pre-production environments to ensure **scalability and efficiency** for handling **Big Data velocities in Production** . **Eliminated Risk of non-compliance with GDPR Policies** that may attract **millions of dollars in fines** if violated.

SKILLS

PROGRAMMING

Python • JavaScript
Go • SQL • Java • C++

FRAMEWORKS

FastAPI • Spark • Kafka
Redis • RabbitMQ
Apache Kafka • NginX
ASGI/WSGI • Hadoop • LangChain
LangGraph • GraphQL
MongoDB • ElasticSearch
Neo4j • Page-Index • RAG
PgVector • BERT • GCP

PLATFORMS

Git • Openshift • Cloudera
Kubernetes • AWS • Docker

EDUCATION

NIT TRICHY

BACHELOR OF TECHNOLOGY

Graduation : May 2022

COURSEWORK

Data Structures And Algorithms
Operating Systems
Database Management Systems
Networks And Protocols
BigData Analytics
Machine Learning and Deep Learning
Software Engineering
Industrial Robotics

ACHIEVEMENTS

1st Position at EYRC, International Robotics Competition (AI and Robotics Competition at IIT Bombay).

Successfully planned and implemented **Negative News Tracking and Risk Score Pipeline** for improving KYC Work Item resolution time. Maintained these respective Data Warehouses for a period of 14 Months overseeing **4 TB of Data** .